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Leveraging Neural Networks In a **Hybrid Model Predictive Control** framework for District Heating Networks

Full paper #87

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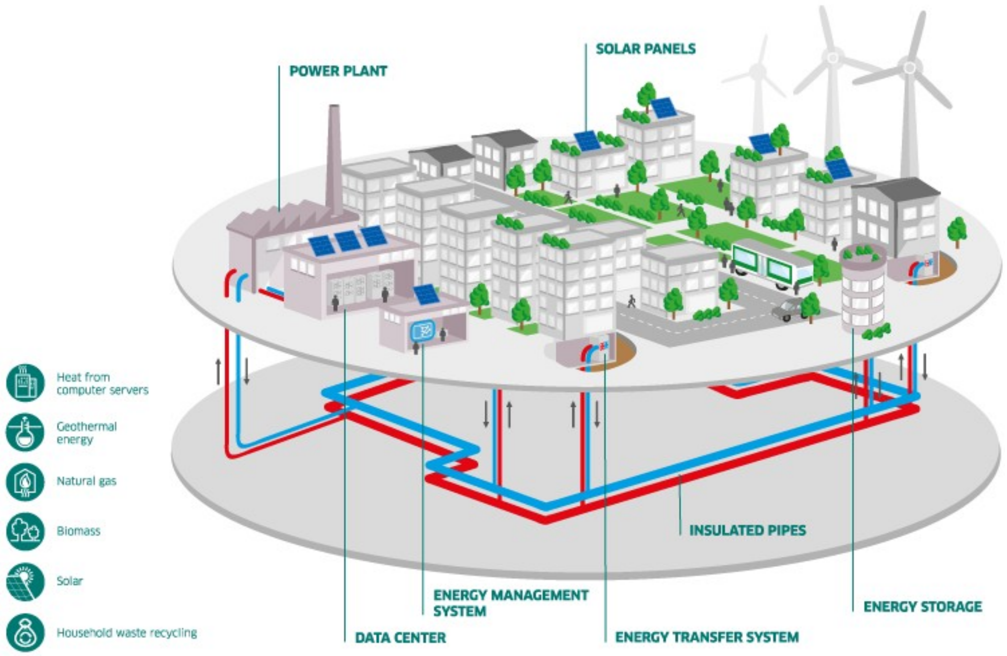
Context



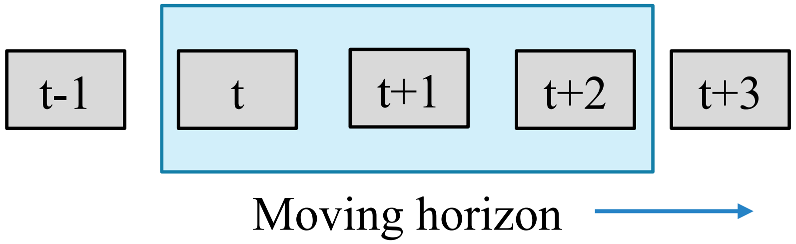
District Heating Network (DHN)
Source: ENGIE website [link]

- A local sized heat energy supply system
- Hot waters traverse **network of pipes** to transport heat from sources to consumers
- Enhances flexibility and efficiency
  - **Optimal control of the network** (Jansen *et al.*, 2023)
  - Integration of renewable sources (Saloux *et al.*, 2023

Model Predictive Control (MPC)

- A subcategory of control optimization
- High effectiveness by combining predictions and decision optimization (Jansen *et al.*, 2023)
- Can be used with analytic or metaheuristics based optimizer

- Use of moving horizon in which decisions are taken (Marquant *et al.*, 2023)



- Optimization within the horizon
- Applying the first step decisions
- Slide the horizon to one step

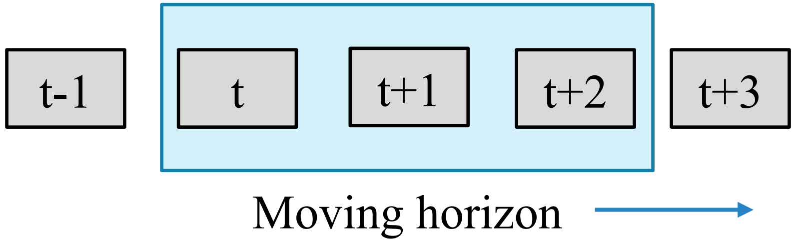
Model Predictive Control (MPC)

<u>In our case:</u>

- Decide the optimal sequence of power to produce, controlled by
  - Sources supply temperatures T_{ss}
  - Supply mass flow rates $\dot{m}_s$

- Optimally fulfilling all consumers heating demands

- Use of moving horizon in which decisions are taken (Marquant *et al.*, 2023)



- Optimization within the horizon
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Model Predictive Control (MPC)

$$J = \sum_{t=1}^{N_t} \left[\sum_{\{i \in S\}} \dot{\boldsymbol{m}}_{s_i}^{\boldsymbol{t}} c_p \left(\boldsymbol{T}_{ss_i}^{\boldsymbol{t}} - T_{sr_i} \right) \right] + M \left(\max\left(0, (\delta_{surplus} - 2)\right)^2 + \max\left(0, (\delta_{deficits} - 2)\right)^2 \right)$$

$$\delta_{deficits} = 100 * \frac{\sum_{(i,t)} \min(\delta_i^t, 0)}{\sum_{(i,t)} D_i^t} \qquad \delta_{surplus} = 100 * \frac{\sum_{(i,t)} \max(\delta_i^t, 0)}{\sum_{(i,t)} D_i^t} \qquad \delta_i^t = D_i^t - \dot{m}_c c_p \left(T_{s_i}^t - T_{rc_i} \right)$$

- Requires the numerical solving of differential equations

(mass balance) $$\forall (i,t),\ \dot{\boldsymbol{m}}_{s_i}^{\boldsymbol{t}} + \sum_{\{e \in E_i^+\}} \dot{m}_e^t = \dot{m}_{c_i} + \sum_{\{i \in E_i^-\}} \dot{m}_e^t \qquad (1)$$

(head loss) $$\forall e, \Delta H_e = \frac{4 f_e l_e u_e^2}{2 g_{rv} d_e} \qquad (2)$$

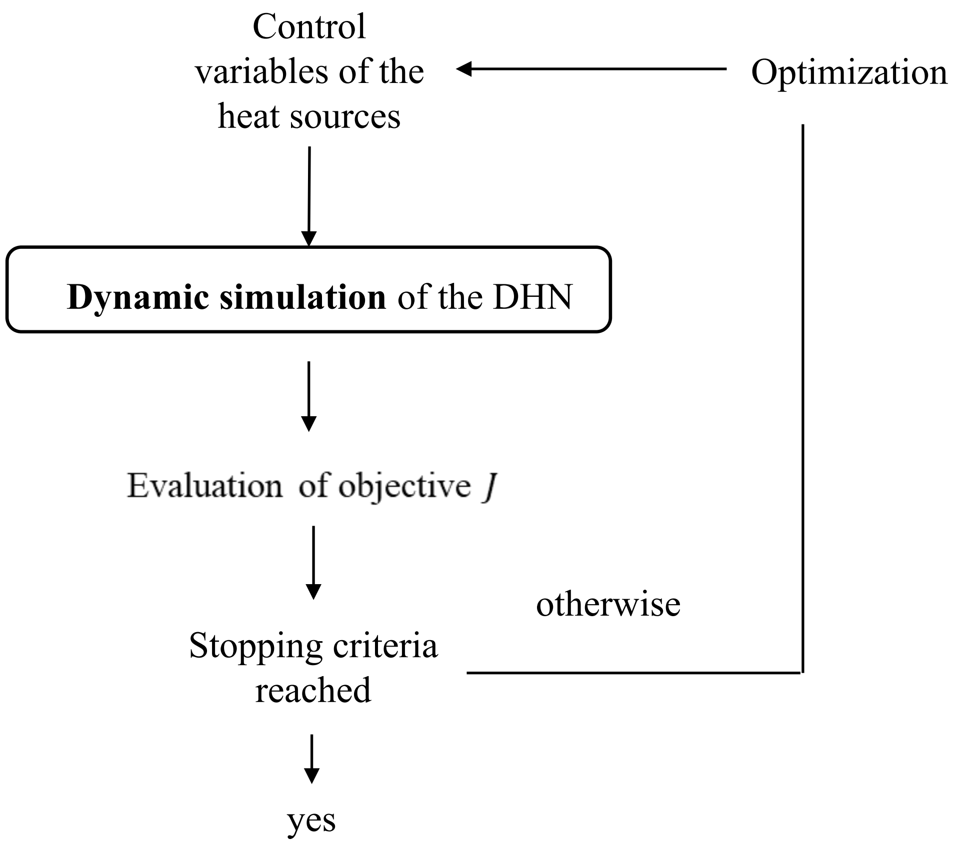
(consumption) $$\forall (i \in C), P_{c_i}^t = \dot{m}_{c_i}^t c_p (T_{s_i} - T_{rc_i}) \qquad (3)$$

(supply balance) $$\forall (i,t), \dot{\boldsymbol{m}}_{s_i}^{\boldsymbol{t}} \boldsymbol{T}_{ss_i}^{\boldsymbol{t}} + \sum_{\{e \in E_i^+\}} \dot{m}_e^t T_{s_i}^t = \left(\dot{\boldsymbol{m}}_{c_i}^{\boldsymbol{t}} + \sum_{\{e \in E_i^-\}} \dot{m}_e^t \right) T_{s_i}^t \qquad (4)$$

(return balance) $$\forall (i,t), \dot{m}_{c_i}^t T_{rc_i}^t + \sum_{\{e \in E_i^-\}} \dot{m}_e^t T_{r_i}^t = \left(\dot{\boldsymbol{m}}_{s_i}^{\boldsymbol{t}} + \sum_{\{e \in E_i^+\}} \dot{m}_e^t \right) T_{r_i}^t \qquad (5)$$

(heat transport) $$\forall e, \frac{\partial T_e}{\partial t} = -\frac{4 \dot{m}_e}{\pi \rho d_e^2} \frac{\partial T_e}{\partial x} - \frac{4 h_e}{\rho c_p d_e} (T_e - T_{ground}) \qquad (6)$$

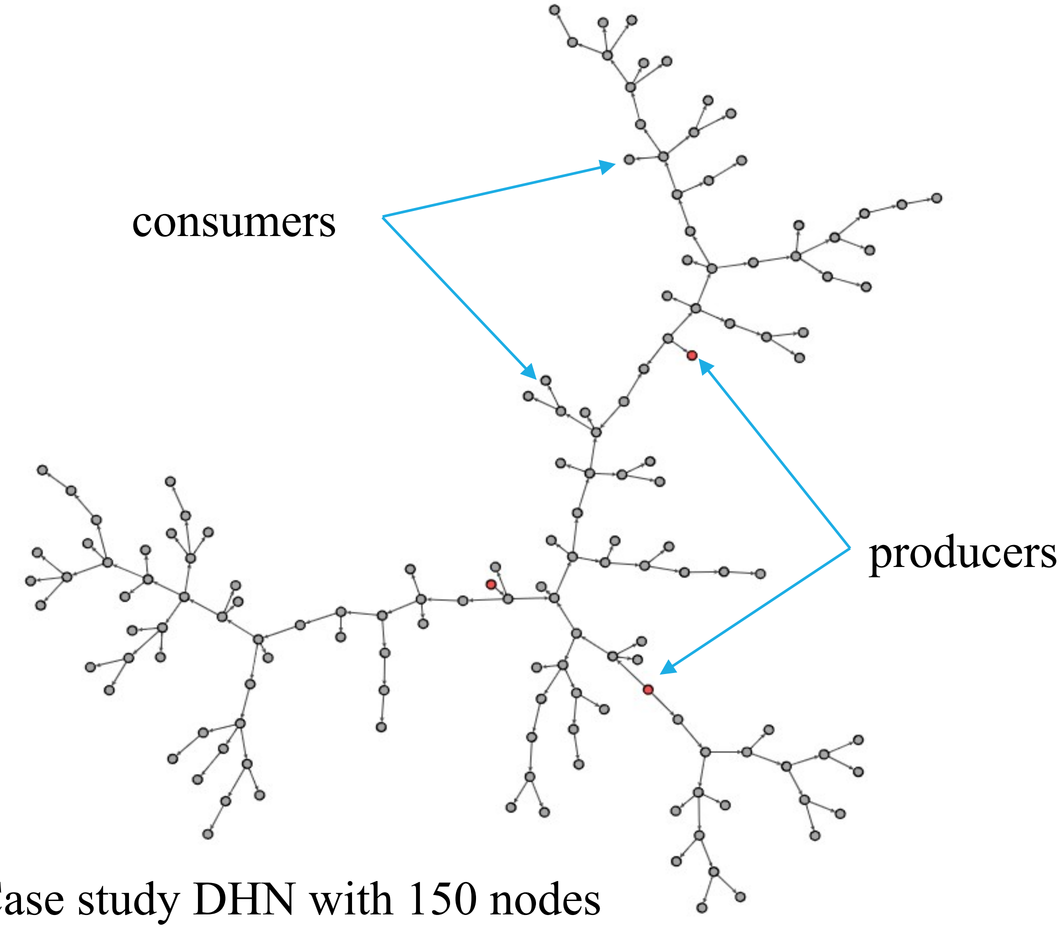
Problematic



Computationally intensive

- A repetitive dynamic simulation of the DHN is required for each time step.
- One simulation is computationally expensive for large DHN.
- **Proposed approach**: reduce the computational costs (CPU time) by reducing the topology of the DHN.

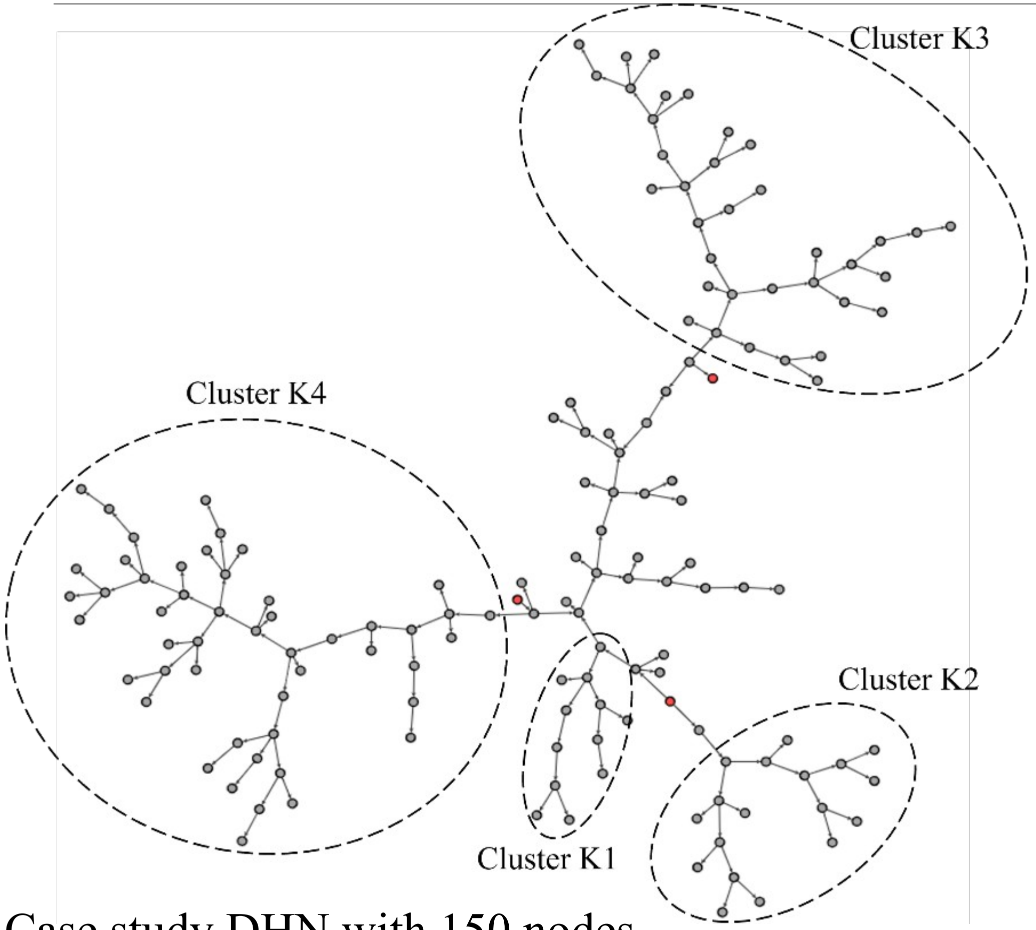
Proposed methodology



Case study DHN with 150 nodes

- DHN viewed as oriented graph

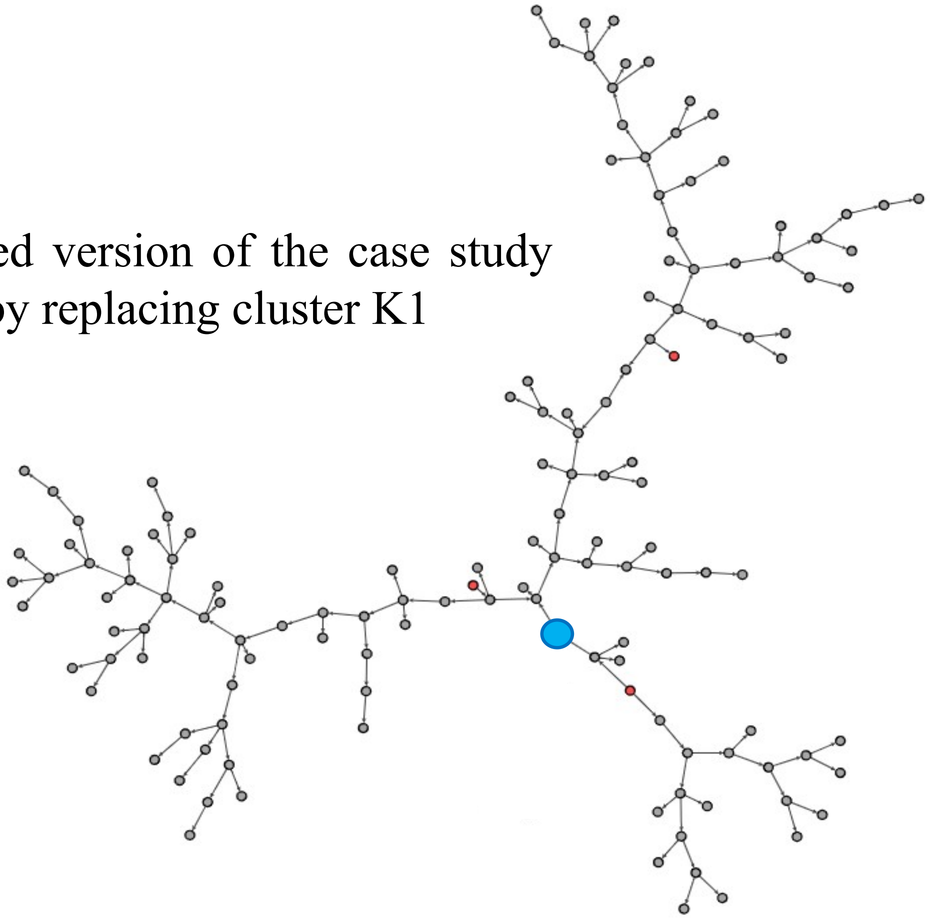
Proposed methodology



Case study DHN with 150 nodes

- DHN viewed as oriented graph
- Identify or **select clusters** of consumer nodes (*i.e.*, K1-K4)

Proposed methodology

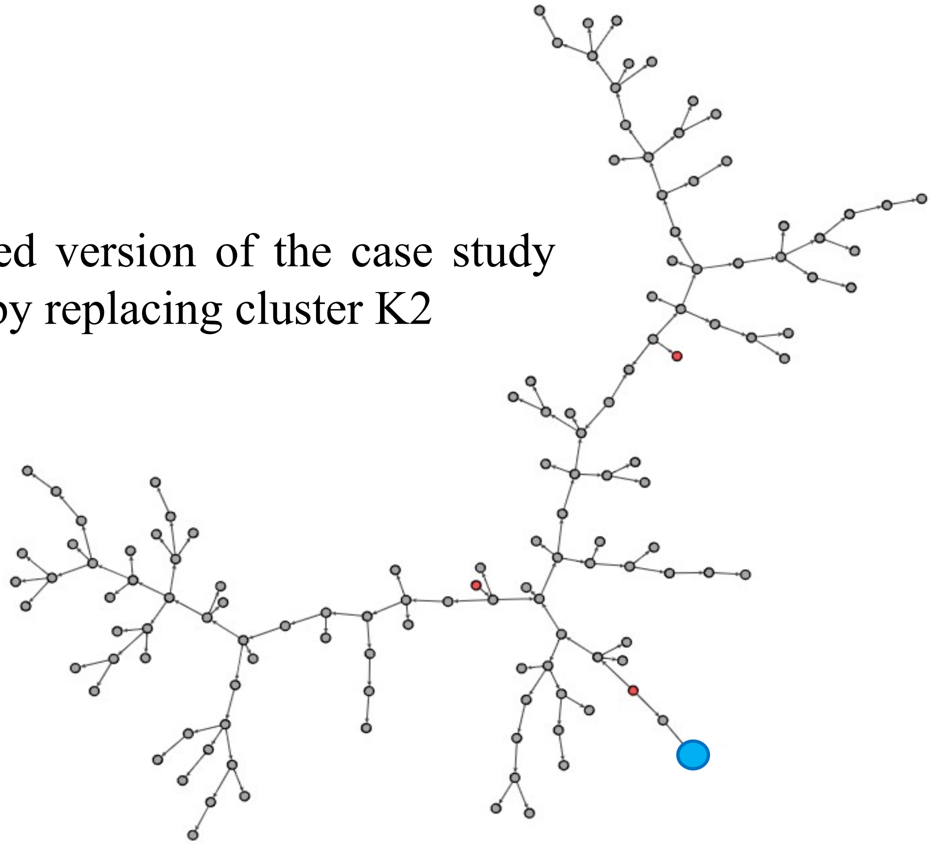


Reduced version of the case study DHN by replacing cluster K1

- DHN viewed as oriented graph
- Identify or select clusters of consumer nodes (*i.e.*, K1-K4)
- Replace each cluster with a trained neural networks acting as surrogate of the cluster

Proposed methodology

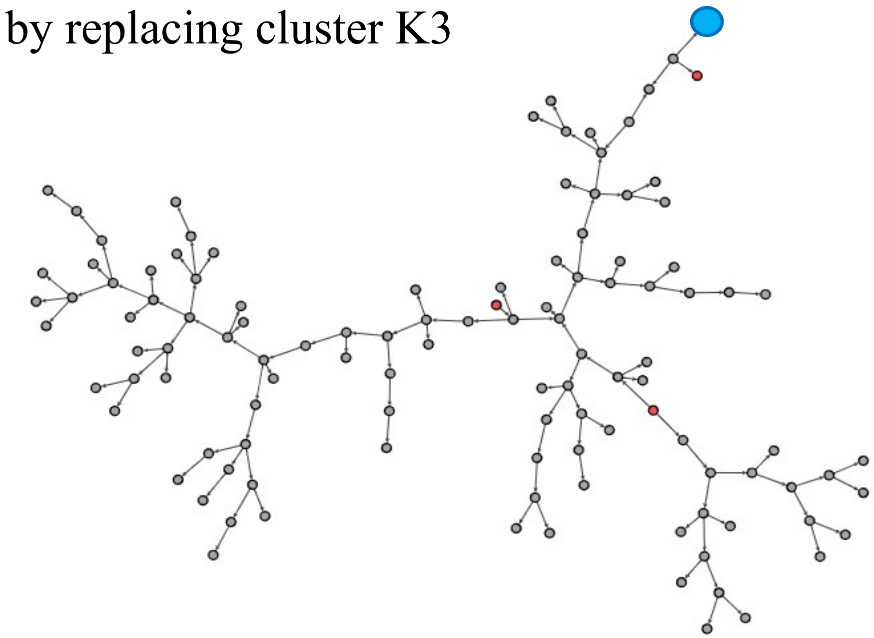
Reduced version of the case study DHN by replacing cluster K2



- DHN viewed as oriented graph
- Identify or select clusters of consumer nodes (*i.e.*, K1-K4)
- Replace each cluster with a trained neural networks acting as surrogate of the cluster

Proposed methodology

Reduced version of the case study DHN by replacing cluster K3



- DHN viewed as oriented graph
- Identify or select clusters of consumer nodes (*i.e.*, K1-K4)
- Replace each cluster with a trained neural networks acting as surrogate of the cluster

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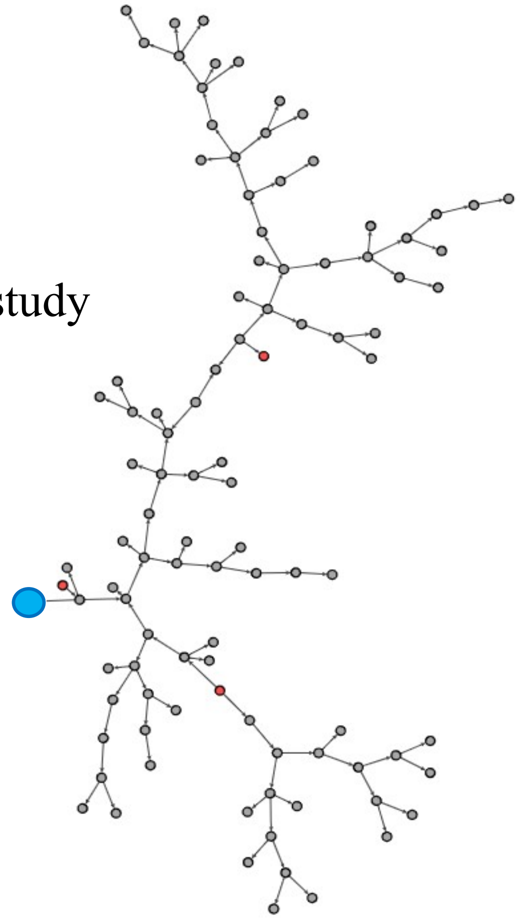
Proposed methodology

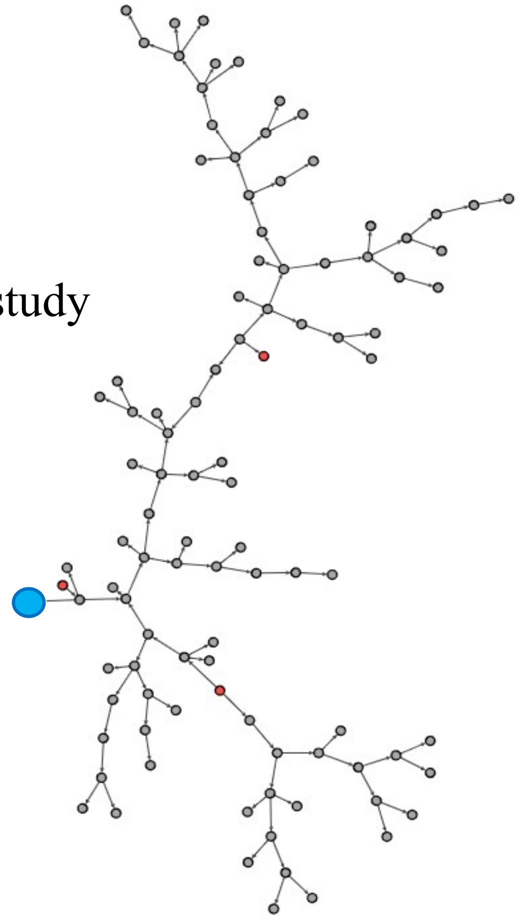
Reduced version of the case study DHN by replacing cluster K4



- DHN viewed as oriented graph
- Identify or select clusters of consumer nodes (*i.e.*, K1-K4)
- Replace each cluster with a trained neural networks acting as surrogate of the cluster

Proposed methodology

Reduced version of the case study DHN by replacing cluster K4



- DHN viewed as oriented graph
- Identify or select clusters of consumer nodes (*i.e.*, K1-K4)
- Replace each cluster with a trained neural networks acting as surrogate of the cluster
  - Maintain temperatures at interfaces
  - Maintain consumptions

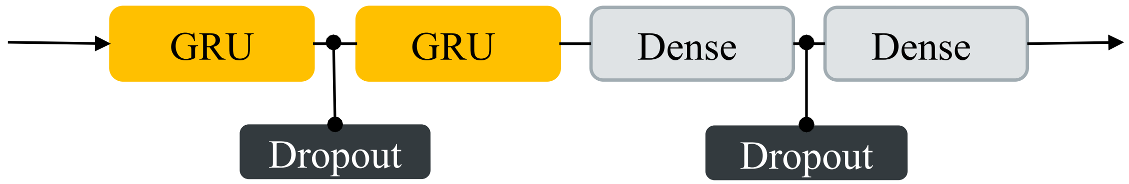
Surrogate neural networks models

- GRU-NN based architectures with hyperparameters selected from grid search
- Data from simulation of the case study DHN (2880 h × 1 minute step)

Incoming water flows
+ Temperatures
+ Mass flow rates

Nodes heating demands
(nodes of replaced cluster)

(Chung *et al.*, 2014)

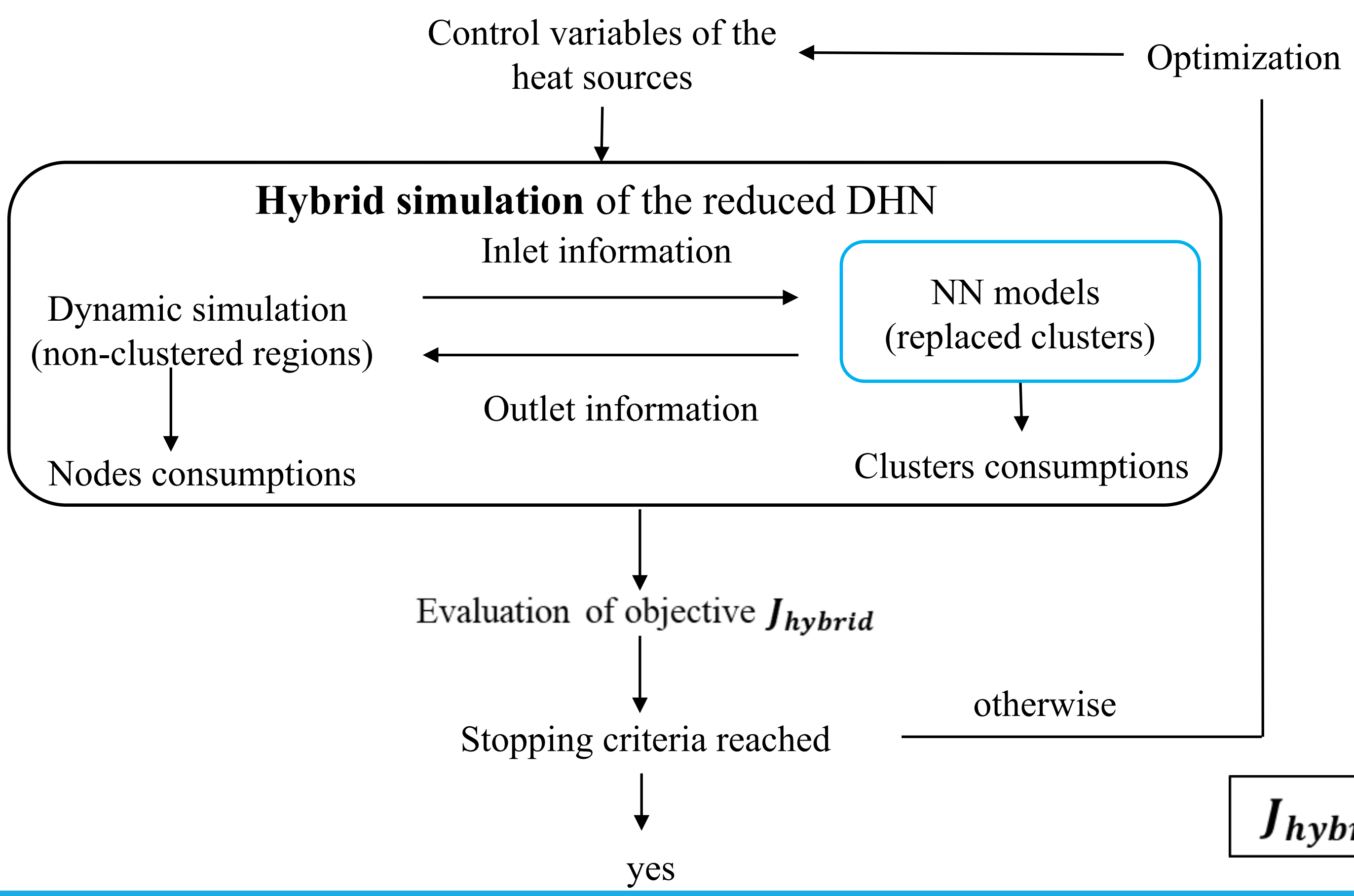


Outgoing water flows
+ Temperatures

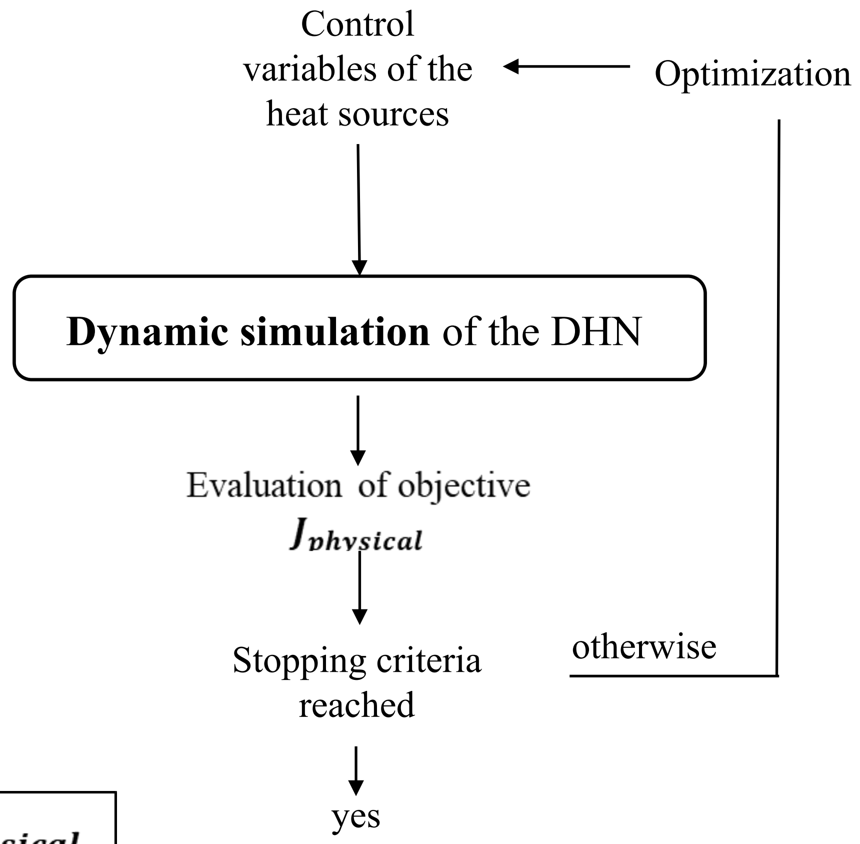
Cluster effective heat consumption

- Training procedures:
  - Temporal split: first 2/3 of data is used for train, and last 1/3 is used for test
  - Adam optimizer (40 epochs, early stopping, decreasing learning rate, mse)

Hybrid MPC vs Physical MPC



$$\boxed{J_{hybrid} \approx J_{physical}}$$



Results (learning performances)

*Nb: number of *MAPE: Mean Absolute Percentage Error

	Cluster K1	**Cluster K2**	**Cluster K3**	**Cluster K4**
Nb clustered nodes	13	18	36	50
Nb output features	3	2	2	2
MAPE (%) on temperatures	0.58	0.06	0.22	0.23
MAPE (%) on consumption ▪ On power level ▪ On energy level	 2.31 0.07	 0.92 0.52	 2.65 0.43	 1.79 0.24

- Ability of the NN models to learn different *sizes* of clusters.
- Lower errors in heat consumption energy than powers as over and under-estimations might cancel off.

Results (Hybrid MPC accuracy)

Results for 10 hours operational running

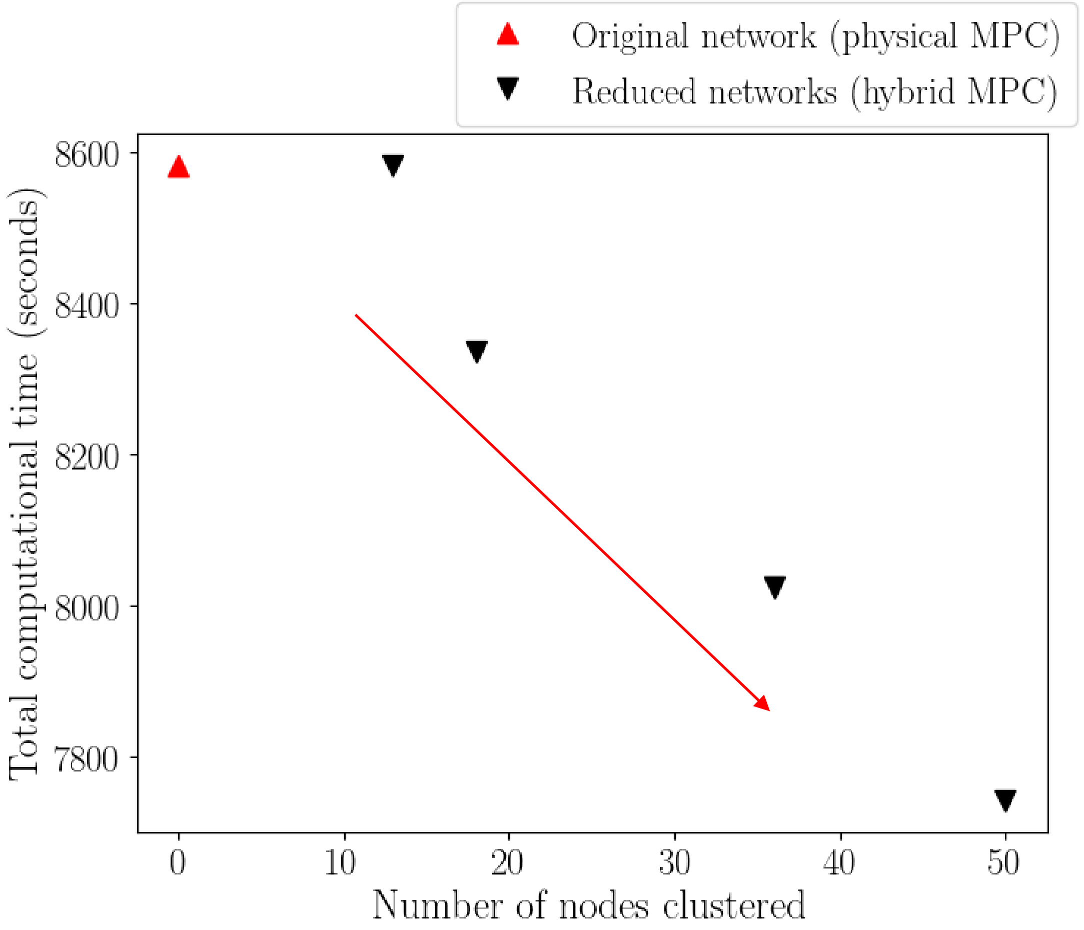
Decisions are taken every dt hour: 1 hour and 0.2 hour (12 minutes).

	Physical MPC	Hybrid MPC				Rule-based (80°C)
Cluster replaced	-	K1	K2	K3	K4	-
dt (hour)						
Produced heat energy (MWh)						

	Physical MPC	Hybrid MPC				Rule-based (80°C)
Cluster replaced	-	K1	K2	K3	K4	-
dt (hour)	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)	1 (0.2)
Produced heat energy (MWh)	**202.8 (218.3)**	**206.6 (203.5)**	**202.7 (218.4)**	**203.2 (213.6)**	**203.6 (209.7)**	**235.7 (232.4)**
$\delta_{deficits}$	1.46 (0.71)	1.37 (0.20)	1.44 (0.68)	1.38 (0.57)	1.40 (0.57)	1.32 (0.35)
$\delta_{surplus}$	0.41 (0.62)	0.68 (0.51)	0.39 (0.63)	0.40 (0.45)	0.44 (0.36)	5.94 (0.95)

- Hybrid MPC has satisfyingly conserved the objective function values

Results (CPU gains)



- Decrease of optimization computational time proportionally to the clusters sizes
- 9 % decrease for largest cluster
- Effective gains come from the repetitive simulation and optimization

Conclusion and perspectives

- We proposed an hybrid MPC approach to optimally control DHNs with lower CPU time
- Surrogate Neural Networks based models learn and replace consumers clusters within the DHNs
- Satisfying accuracy and computational time reductions validate our approach

!! Further reduction of CPU times will be considered

Thank you for your attention

Questions ?